

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I K.AISHWARYA(192572116) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Given

- One **4-gallon jug**
- One **3-gallon jug**
- No measuring marks
- Unlimited water from the pump
- Water can be poured between jugs or onto the ground
- **Goal:** Get exactly **2 gallons in the 4-gallon jug**

Step 1: Represent the state

We represent the amount of water in the two jugs as:

(x, y)

where:

- x = water in the 4-gallon jug
- y = water in the 3-gallon jug

Initially:

(0, 0)

Step 2: Find a sequence of actions

Think about how to create exactly **2 gallons**.

1. Fill the **3-gallon jug** $\rightarrow (0, 3)$
2. Pour the 3-gallon jug into the 4-gallon jug $\rightarrow (3, 0)$
3. Fill the 3-gallon jug again $\rightarrow (3, 3)$
4. Pour from the 3-gallon jug into the 4-gallon jug.
The 4-gallon jug already contains 3 gallons, so it can accept only **1 gallon**.
Therefore, **2 gallons remain in the 3-gallon jug** $\rightarrow (4, 2)$
5. Empty the 4-gallon jug $\rightarrow (0, 2)$
6. Pour the remaining 2 gallons into the 4-gallon jug $\rightarrow (2, 0)$

State sequence

(0,0)
 ↓ Fill 3G
(0,3)
 ↓ Pour 3G → 4G
(3,0)
 ↓ Fill 3G
(3,3)
 ↓ Pour 3G → 4G
(4,2)
 ↓ Empty 4G
(0,2)
 ↓ Pour 3G → 4G
(2,0)

Final state

(2,0)

Therefore, **exactly 2 gallons of water are obtained in the 4-gallon jug.**

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?

Percepts are the information received by the agent through its sensors.

For a Mars Rover, percepts include:

- Images of the Martian surface
- Rocks and soil
- Temperature
- Atmospheric conditions
- Obstacles
- Rover's current location
- Chemical composition of samples
- Signs of water or minerals
- Battery/power level

- ii. Characterize the operating environment.

	Mars Rover
Observability	Partially observable
Agents	Single-agent
Determinism	Stochastic

	Mars Rover
Episodes	Sequential
Environment changes	Dynamic
State space	Continuous
Knowledge	Partially unknown

Why?

- **Partially observable:** The rover cannot observe the entire planet at once.
- **Single-agent:** The rover primarily makes its own decisions.
- **Stochastic:** Outcomes can be affected by uncertain terrain and environmental conditions.
- **Sequential:** One action can affect future actions.
- **Dynamic:** Weather, dust, temperature, and other conditions can change.
- **Continuous:** Location, distance, temperature, etc. can have continuous values.
- **Unknown:** The rover does not initially know everything about Mars.

iii. What are the actions the agent can take?

The Mars Rover can:

1. Move forward.
2. Turn left or right.
3. Stop.
4. Take photographs.
5. Collect rock and soil samples.
6. Analyze samples.
7. Detect and avoid obstacles.
8. Send information to Earth.
9. Choose locations for exploration.
10. Manage its energy and return to a safe location when necessary

iv. How can one evaluate the performance of the agent?

We can measure the rover's performance using:

- Area of Mars explored
- Number of useful samples collected
- Accuracy of sample analysis
- Scientific information discovered
- Successful obstacle avoidance
- Amount of useful data transmitted
- Energy efficiency
- Rover safety
- Mission completion

- v. What sort of agent architecture do you think is most suitable for this agent? Why?

A **Model-Based Utility-Based Learning Agent** would be suitable.

Why?

Model-based:

The rover needs an internal model of its surroundings because it cannot observe everything directly.

Utility-based:

It must choose between different actions by considering factors such as scientific value and time.

Learning:

The rover can learn from previous observations and improve its decisions during the mission.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

The objective is to place **8 queens on an 8×8 chessboard** such that **no two queens attack each other**.

A queen can attack another queen if they are in the same:

- Row
- Column
- Diagonal

1. Initial State

- The chessboard is empty.
- Initial state = **No queens placed on the board.**

2. State

- A state represents a particular arrangement of queens on the chessboard.
- For example:
- [1, 5, 8]
- can represent queens placed in the first three columns at rows 1, 5, and 8.

3. Actions

- Place a queen in an empty position where it does not attack any previously placed queen.

4. Transition Model

- When a valid queen is placed, the system moves from the current configuration to a new configuration containing one additional queen.

5. Goal Test

The goal is reached when:

- All **8 queens** are placed.
- No two queens attack each other.

6. Path Cost

Each queen placement can be considered to have equal cost.

Therefore: Path cost = number of queens placed.

Search Strategy

A suitable strategy is **Depth-First Search (DFS) with Backtracking**.

How it works

1. Start with an empty board.
2. Place a queen in the first column.
3. Move to the next column.
4. Check possible rows.
5. If a safe position is found, place the queen.
6. Continue to the next column.
7. If no safe position is available, **backtrack**.
8. Remove the previously placed queen.
9. Try another position.
10. Continue until all 8 queens are successfully placed.

Start



Place Queen



Is position safe?



Yes → Place next Queen



Continue



8 Queens placed?



Yes → Solution



No



Backtrack



Try another position

One valid solution

A valid arrangement is:

[1, 5, 8, 6, 3, 7, 2, 4]

This means:

Column	Row
1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

No two queens share the same row, column, or diagonal.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Problem

A customer wants to travel from a **source location to a destination** using a cab-booking application. The customer can choose different cab types such as:

- Micro
- Mini
- Sedan
- Prime
- Shared

The system should help the customer select a suitable cab.

Type of Problem-Solving Agent

The most suitable agent is a **Utility-Based Agent**.

Why?

A utility-based agent compares different choices based on several factors and selects the option that provides the **best overall benefit**.

For cab booking, it can consider:

- Cab fare
- Travel time
- Distance
- Cab availability
- Customer comfort
- Cab type
- Customer preference
- Traffic conditions

So, instead of simply selecting the first available cab, it chooses the cab with the **highest utility**.

Problem Formulation

Initial State:

Customer is at the pickup location.

Goal State:

Customer reaches the destination.

Actions:

1. Enter pickup location.
2. Enter destination.
3. Select preferred cab type.
4. Find available cabs.
5. Compare available options.
6. Book the selected cab.
7. Travel to destination.

Performance Measure:

The agent should try to:

Minimize cost and travel time while maximizing comfort, availability, and customer satisfaction.

Pseudocode

START

Input pickup location

Input destination

Input customer preference

Find available cabs

For each available cab:

 Calculate fare

 Calculate estimated travel time

 Check cab type

 Check comfort

 Check customer preference

Calculate utility of each cab

Select cab with highest utility

Book the selected cab

Track the cab

Start journey

Track route until destination is reached

Display trip details and fare

STOP

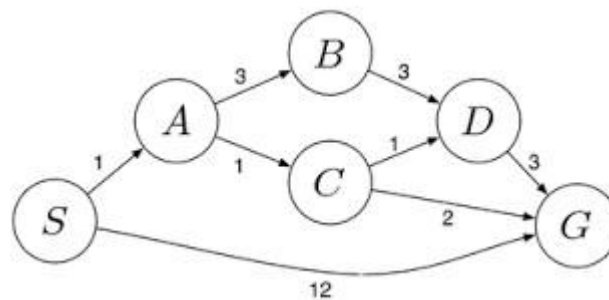
Final Answer

Type of agent: Utility-Based Agent

Reason: It evaluates multiple factors such as cost, time, comfort, availability, and customer preference before selecting the best cab.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Given graph

The route costs are:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Initial state: S

Goal state: G

UCS always expands the node with the **lowest cumulative path cost**.

Step-by-step UCS

Step 1: Start at S

Cost of S = **0**

From S:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$

Priority queue:

Node	Cost
A	1
G	12

Choose **A**, because it has the lowest cost.

Step 2: Expand A

A has cost **1**.

From A:

- $A \rightarrow B = 3 \rightarrow \text{total cost} = 1 + 3 = 4$
- $A \rightarrow C = 1 \rightarrow \text{total cost} = 1 + 1 = 2$

Priority queue:

Node	Cost
C	2
B	4
G	12

Step 3: Expand C

C has cost **2**.

From C:

- $C \rightarrow D = 1 \rightarrow \text{total} = 2 + 1 = 3$
- $C \rightarrow G = 2 \rightarrow \text{total} = 2 + 2 = 4$

Priority queue:

Node	Cost
D	3
B	4
G	4

Node	Cost
G	12

Choose **D**, because its cost is 3.

Step 4: Expand D

D has cost 3.

From D:

- $D \rightarrow G = 3$

Therefore:

$$3+3=6$$

So another path to G has cost **6**.

But we already have a path to G with cost **4**.

Priority queue:

Node	Cost
B	4
G	4
G	6
G	12

Choose **G** with cost 4.

Since G is the goal, **UCS stops**.

Final Answer

Least-cost path:

$$S \rightarrow A \rightarrow C \rightarrow G$$

Total cost:

$$1+1+2=4$$

UCS Expansion Order

$$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$$

Final statement for assessment

Using Uniform Cost Search, the least-cost path from the starting warehouse S to the destination G is $S \rightarrow A \rightarrow C \rightarrow G$, with a minimum total cost of 4.

